



Blending of NR/BR/ENR/EPDM-g-GMA by reactive processing for tire sidewall applications: Effects of grafting and ENR on curing characteristics, mechanical properties, and dynamic ozone resistance

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Abstract

Application of ethylene-propylene-diene monomer (EPDM) grafted with glycidyl methacrylate (GMA), along with epoxidized natural rubber (ENR) was investigated for tire sidewall blend. The presence of ENR and EPDM-g-GMA improved physico-mechanical properties of the blends. The best results were obtained using 5 phr EPDM-g-GMA in the blend. Tensile and tear strength of the blend with 5 phr EPDM-g-GMA were 13.7% and 46.5% higher than conventional blend of tire sidewall, respectively. Ozone resistance in dynamic condition of NR/BR/ENR with 5 phr EPDM-g-GMA were 20% and 30% superior than NR/butadiene rubber (BR)/ENR/EPDM and conventional NR/BR with *N*-(1,3-dimethylbutyl)-*N'*-phenyl-*p*-phenylene, respectively. Improved mechanical and ozone resistance properties are due to better homogeneity of cross-link density and carbon black distribution in the blend.

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Keywords

Ozone resistance, natural rubber, butadiene rubber, glycidyl methacrylate, epoxidized natural rubber, tire sidewall, tear strength

Introduction

The natural rubber/butadiene rubber (NR/BR) blend in tire sidewall formulation contains amine compounds such as *N*-(1,3-dimethylbutyl)-*N'*-phenyl-*p*-phenylene diamine, (6PPD), as antiozonant. This antiozonant may migrate to the surface and stain the tire sidewall. Several methods for synthesizing nonstaining antiozonants have been tested to overcome this issue, though with little success. Alternatively, rubbers with inherent ozone resistance such as ethylene-propylene-diene monomer (EPDM) and halo butyls can be used in the NR/BR blend. However, there are three main challenges with the use of rubber in the blend: firstly, the two highly saturated rubbers may co-cure with other's unsaturated rubbers.¹ NR and BR are rubbers with low percentage of saturation and concentration of carbon double bond (C=C). Secondly, the level of polarity and curing reaction sites are different from EPDM.² Finally, as carbon black has a tendency to rubbers with lower saturation, there will be an inhomogeneous distribution of filler in the blend.³ Functionalized EPDM with *N*-chlorothio-*N*-butylbenzen sulfonamide, or maleic anhydride have been used to reduce the co-curing and distribution homogeneity properties. Zhang et al.⁴ used ethylene-propylene monomer (EPM) grafted with maleic anhydride in the blend of NR/BR/EPDM. They reported that ozone resistance in static and dynamic condition in the presence of 30 phr of EPDM with ethylene-para-methyl-styrene were comparable with blends of NR/BR containing 6PPD. In all similar suggested procedures, attempts were made to increase the curing rate of EPDM.⁴⁻¹¹

Sahakaro et al.⁵ investigated the ozone resistance of NR/BR/EPDM and NR/BR blend in two states of conventional straight mixing, and reactive processing techniques. They grafted a common *N*-cyclohexyl-2-benzothiazole sulfonamide (CBS) onto EPDM prior to blending with NR/BR, to overcome the cure incompatibility in NR/BR/EPDM blends. The reactive blends showed a significant improvement in tensile properties and blend morphology. The increase of tensile properties in reactive blends suggests homogeneous blend. The properties were equal or superior to those of the conventional NR/BR tire sidewall compounds. They reported that using 30 phr EPDM improves ozone resistance of the blend to the required level, without the need for 6PPD.⁵⁻⁷

The main objective of this research is to replace 6PPD antiozonant with EPDM-glycidyl methacrylate (GMA) to improve the ozone resistance of conventional tire sidewall formulation. The novelty of this work relies on using EPDM grafted with GMA and using ENR in tire sidewall formulation. Tensile and tear properties and dynamic ozone resistance of NR/BR/ENR/EPDM-g-GMA blend with three levels of EPDM-g-GMA, NR/BR/ENR/EPDM blend, and NR/BR blend with 6PPD were determined to investigate the optimum level of EPDM-g-GMA in compound.

Experimental

Materials

Natural rubber (Standard Malaysian Rubber, SMR-20, GP grade), high cis-BR (BR-1220, Arak Petrochemical Co., Iran), epoxidized natural rubber (ENR-50, Kumpulan Guthrie Berhad Co., Malaysia) and EPDM containing 4.5% ethylene diene norbornene as third monomer and 52% ethylene (Kep-270, Kumho, South Korea) were basic rubbers used. GMA with density of 1.07 g cm^{-3} and molecular weight of $142.15 \text{ g mol}^{-1}$ used for grafting onto EPDM was supplied by Merck Co., Germany. Other materials used were xylene, acetone, potassium hydroxide (KOH), 3-chloro acetic acid, and ethanol from Merck Co., dicumyl peroxide (DCP), CBS, 2,2,4-Trimethyl-1,2-dihydroquinoline (TMQ), and 6PPD from Rochem Co., China, stearic acid from Palma Co., Malaysia, zinc oxide from Totia Co., Iran, sulfur from Testak Co., Iran, oil-840 from Behran Co., Iran, and carbon black-N330 from Pars Co., Iran.

Grafting of GMA on EPDM

Grafting of EPDM was carried out using a Brabender laboratory mixer (model 50E; Germany) with the capacity of 55 mL. Initial conditions of 110°C and rotor speed of 35 r min^{-1} , and final conditions of 130°C and 55 r min^{-1} were chosen according to the procedure in Ref. 12. EPDM was premixed for 2 min. Then a homogenized solution containing 2 phr DCP and 5, 10, and 30 phr GMA was added. Mixing was carried out for 8 min.¹²

Preparation and vulcanization of compounds

Five different compounds were prepared according to Table 1 using a Brabender internal mixer with the volumetric capacity of 1500 mL and rotor speed of 60 r min^{-1} . Initially, NR was added to a mixing chamber and premixed for 1 min. ENR, BR, and then EPDM or EPDM-g-GMA were subsequently added to the mixing chamber. Carbon black and aromatic oil (oil-840) were added after 2 min of mixing. When the chamber temperature reached 105°C , other ingredients including stearic acid, zinc oxide, and TMQ were added. The prepared compound was then milled in a two-roll mill for 2–3 min. In the final stage of mixing in the internal mixer, curing agents including sulfur and CBS were added to compounds to reach 70°C , and then milled and cooled at room temperature.

Sample characterizations

Confirmation and amount of GMA grafted onto EPDM was investigated using Fourier transform infrared (FTIR) spectroscopy and titration, respectively. The EPDM-g-GMA samples (2 g) were dissolved in xylene at $150\text{--}200^\circ\text{C}$. The solution was poured in 200 mL acetone:ethanol (1:1), as an antisolvent, and stirred for 15 min. Then the samples were filtered and washed by acetone using Soxhlet method to extract any unreacted GMA. Finally, samples were dried in a vacuum oven at 80°C for 12 h. The FTIR spectra of dried samples were acquired by a Thermo Fisher Scientific model spectrometer (Waltham, Massachusetts, USA), in the spectrum of $4000\text{--}400 \text{ cm}^{-1}$.

Table 1. Formulation of the compounds.

Components	Compound code				
	Ref-0	Ref-I	GMA-5	GMA-10	GMA-30
NR	50	26.25	26.25	26.25	26.25
BR	50	35	35	35	35
ENR	—	8.75	8.75	8.75	8.75
EPDM	—	30	25	20	—
EPDM-g-GMA	—	—	5	10	30
Zinc oxide	4	4	4	4	4
Stearic acid	2	2	2	2	2
TMQ	2	2	2	2	2
CBS	2	2	2	2	2
Sulfur	2	2	2	2	2
CB-N330	50	50	50	50	50
Oil-840	10	10	10	10	10
6PPD	2	—	—	—	—

NR: natural rubber; BR: butadiene rubber; ENR: epoxidized natural rubber; EPDM: ethylene-propylene-diene monomer; EPDM-g-GMA: ethylene-propylene-diene monomer-grafted glycidyl methacrylate; CBS: *N*-cyclohexyl-2-benzothiazole sulfonamide; CB-N330: carbon black-N330; 6PPD: *N*-(1,3-dimethylbutyl)-*N'*-phenyl-*p*-phenylene.

The amount of grafted GMA grafted onto EPDM was investigated using a back-titration method.¹³ The dried sample (0.5 g) was dissolved in xylene at 150°C, and then 3-chloro acetic acid solution was added. The solution was kept in an oven at $130 \pm 20^\circ\text{C}$ for 90 min to complete the reaction of the acid group with epoxy group of GMA. The polymer was extracted by adding acetone and ethanol and then the solution was filtered. The filtrate was titrated with 0.1 N KOH solution and the percent of grafted GMA was calculated using equation 1.

$$\% \text{GMA} = \frac{142.16(V_{A0} - V_A) \times N_A}{W}$$

where N_A , V_{A0} , and V_A are the normality, the initial volume, and the neutralized volume of 3-chloro acetic acid solution and W is the sample weight.¹³

The optimum curing characteristics of compounds were determined using a RPA 2000 cure meter (Alpha Technology Co., USA), according to ISO 6502.

Tensile tests were carried out using Zwick tensometer according to ASTM D-412. Test samples were prepared from compression-molded sheets as dumb-bell type specimens.

Trouser-type specimens were prepared from compression-molded sheets and tear test was carried out according to DIN 53615.

Dynamic ozone resistance of compounds was measured according to ISO 143-1. The test was carried out at an ozone concentration of 100 ± 5 pphm (parts per hundred million), elongation of 20% at $40 \pm 2^\circ\text{C}$ and relative humidity of $50 \pm 5\%$ for 100 h.

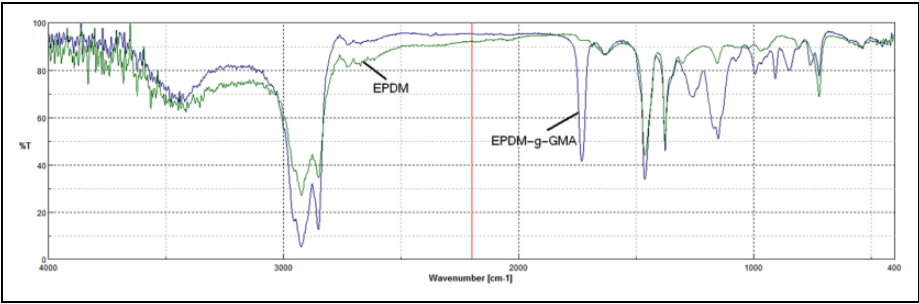


Figure 1. FTIR spectra of EPDM-g-GMA in comparison with that of EPDM. FTIR: Fourier transform infrared; EPDM-g-GMA: ethylene–propylene–diene monomer-grafted glycidyl methacrylate.

Table 2. Infrared absorption peaks and their assignments.

Functional group	Infrared absorption peak (cm ⁻¹)
C–H stretching	2925, 2843
C=O stretching	1730
C–H bending (scissoring)	1462, 1378
C–H out-of plane bending	720
C–S stretching, sulfides (weak)	710–570

Results and discussion

Grafting of EPDM

FTIR spectra of EPDM-g-GMA after purification, in comparison with the spectrum of pure EPDM, are presented in Figure 1. The important functional groups and their assignments are listed in Table 2. For grafted EPDM, the characteristic signal of C=O bond is observed at 1730 cm⁻¹. As solvent extraction was performed on the samples prior to the FTIR test, the clear presence of the absorption peak indicates chemical linkage of GMA to the EPDM molecular chains. FTIR and titration results confirm grafting of EPDM with 6% GMA.

Curing characteristics

Table 3 represents curing characteristics of compounds. Compound coded as Ref-0 shows the highest cure time (*t*_{c,90}) and lowest scorch time (*t*_{s2}).The observation is due to presence of elastomers with double bond in the compound (NR/BR), which has the highest cure rate. Adding EPDM and EPDM-g-GMA to the compound decreases cure time and increases scorch time of the compound, because of the low cure rate of EPDM.

Table 3. Curing characteristics of the compounds.

Curing characteristics	Compounds code				
	Ref-0	Ref-1	GMA-5	GMA-10	GMA-30
Minimum torque (dN m)	0.61	0.66	0.72	0.77	0.79
Maximum torque (dN m)	9.21	9.08	9.32	9.41	9.52
$F_{\max} - F_L$ (dN m)	8.66	8.42	8.60	8.64	8.73
t_{s2} (min:s)	3:14	3:26	3:20	3:17	3:15
$t_{c,90}$ (min:s)	8:15	8:13	7:41	7:31	7:29

t_{s2} : scorch time; t_c : cure time; GMA: glycidyl methacrylate; $F_{\max}-F_L$: the difference between maximum and minimum torque.

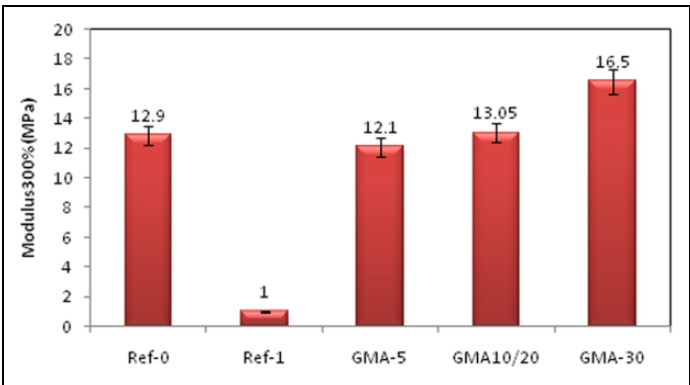


Figure 2. Tensile modulus of compounds at 300% strain.

Tensile properties

Figure 2 represents tensile modulus at 300% of vulcanized blends. Compound-coded as Ref-1 shows the lowest modulus. Tensile modulus increased with increasing the amount of EPDM-g-GMA from 5 phr to 30 phr.

Figure 3 represents tensile strength of the vulcanized blends in comparison with that of compounds containing EPDM grafted with alkyl phenol polysulfide (EPDM-g-APPS) reported by Zhang et al.¹⁴ Compound Ref-1 shows the lowest tensile strength. Tensile strength reaches its maximum value in sample GMA-5. The enhancement is associated with more uniform distribution of carbon black, as well as improved cross-linking across the blend. Increasing the amount of EPDM-g-GMA in the blends reduces the tensile strength, indicating heterogeneous structure of the blends.^{5,14} Compounds with EPDM-g-GMA investigated in this research show higher tensile strength (5%) than compounds with EPDM-g-APPS.

Figure 4 shows elongation at break of vulcanized compounds. Compound Ref-1 shows the lowest elongation at break. Presence of EPDM-g-GMA increases the elongation at break of compounds; however, increasing the amount of EPDM-g-GMA from

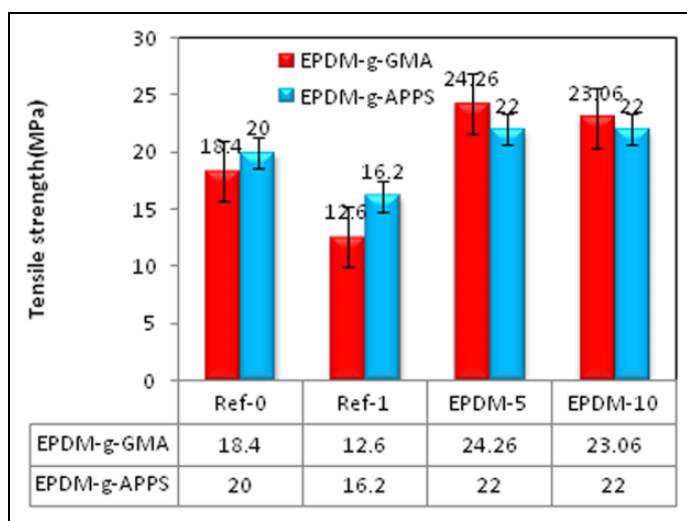


Figure 3. Tensile strength of compounds containing EPDM-g-APPS¹⁴ and EPDM-g-GMA. EPDM-g-APPS: ethylene-propylene-diene monomer-grafted alkyl phenol polysulfide; EPDM-g-GMA: ethylene-propylene-diene monomer-grafted glycidyl methacrylate.

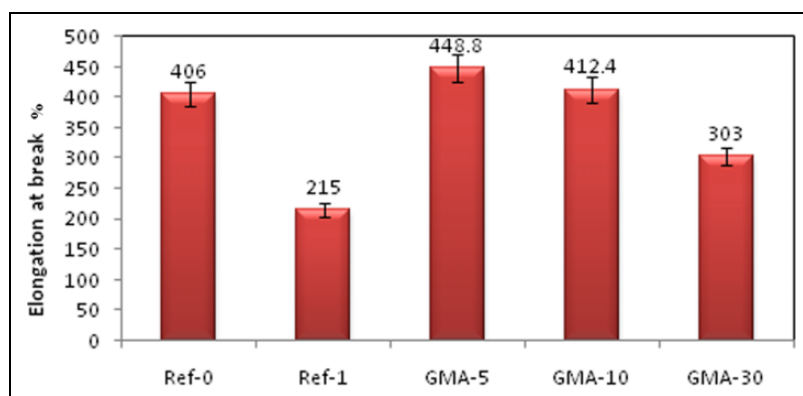


Figure 4. Elongation at break of compounds.

5 phr to 30 phr reduces the elongation at break of compounds. The decrease can be due to in homogeneity of mixing and cross-linking, which relates to increasing heterogeneity in the blend structure.

Tear strength

Figure 5 represents tear strength of trouser test specimens of the compounds in comparison with that of compounds containing EPDM-g-APPS done by Zhang et al.¹⁴ Tear

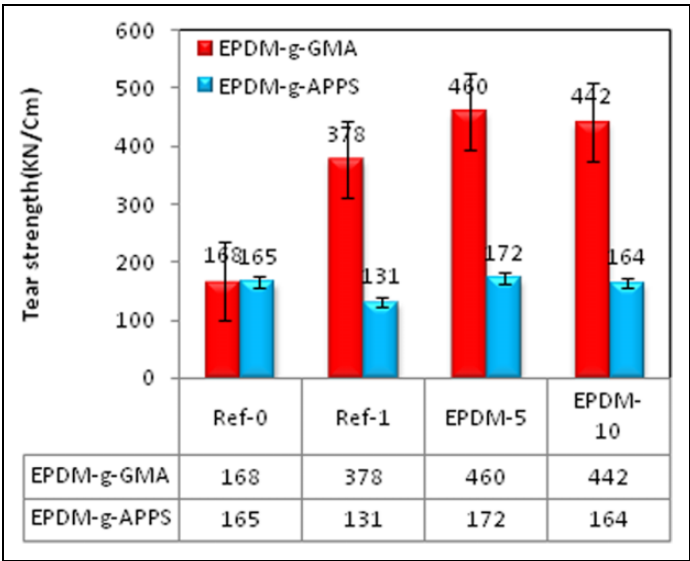


Figure 5. Tear resistance of compounds containing EPDM-g-APPS¹⁴ and EPDM-g-GMA. EPDM-g-APPS: ethylene–propylene–diene monomer-grafted alkyl phenol polysulfide; EPDM-g-GMA: ethylene–propylene–diene monomer-grafted glycidyl methacrylate.

strength of compounds increases significantly with increasing the amount of EPDM-g-GMA to 10 phr, while further increasing the EPDM-g-GMA to 30 phr decreases the tear strength. The decrease is due to lower tear strength of EPDM, as well as poor distribution and lower cross-link density.¹⁴ Compounds with EPDM-g-GMA investigated in this research show 36% higher tear strength than those with EPDM-g-APPS.

Dynamic ozone resistance

Dynamic ozone tests were carried out to compare the resistance to ozone cracking of the conventional NR/BR blends with 6PPD, NR/BR/ENR/EPDM blends, and NR/BR/ENR/EPDM-g-GMA blends.

Some cracks were observed on the surface of compound coded Ref-0 after 48 h. Surface discoloration and staining, caused by the use of 6PPD, were also observed. The observation shows the poor protective properties of 6PPD against ozone attack. Very small cracks were observed on the surface of compound Ref-1 after 23 h, which shows the incompatibility of the rubbers.

No cracks were observed on the surface of compounds coded GMA-5 and GMA-10 after 100 h. This observation is due to the presence of epoxy groups of GMA and ENR that react with ozone and produce ketone compounds. Ketone groups provide good ozone resistance in samples.¹⁵ The image of the GMA-10 sample exposed to ozone after 100 h is shown in Figure 6.

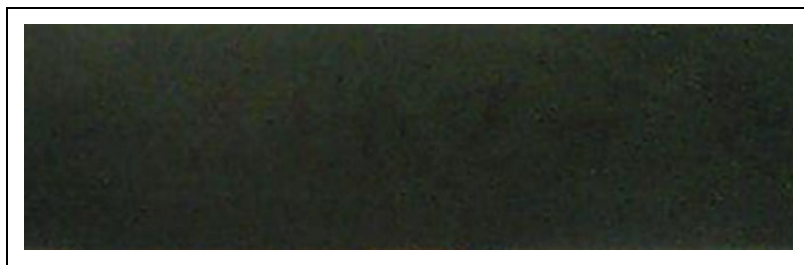


Figure 6. Photographs of the GMA-10 compound after 100 h ozone exposure. GMA: glycidyl methacrylate.



Figure 7. Photographs of the GMA-30 compound after 48 h ozone exposure. GMA: glycidyl methacrylate.

Compound coded GMA-30 cracked after 18 h exposure to ozone, and tore off after 90 h. The reason is decreasing homogeneity in the blend structure with increasing the amount of EPDM-g-GMA to 30 phr.^{1,7} Figure 7 shows the GMA-30 compound after 48 h ozone exposure under testing conditions.

Conclusions

A suitable replacement candidate for conventional sidewall tire blend was found using 5 phr EPDM-g-GMA in NR/BR/ENR blend. Blends of NR/BR/ENR/EPDM-g-GMA with different amounts of EPDM-g-GMA (5, 10, and 30 phr), NR/BR/ENR/EPDM, and conventional NR/BR with 2 phr 6PPD were prepared using reactive processing technique. Effects of grafting and ENR on curing characteristics, mechanical properties, and dynamic ozone resistance of the three blends were measured and compared. Presence of ENR and EPDM-g-GMA improved physico-mechanical properties of the blends. The optimum results were obtained when using 5 phr EPDM-g-GMA in the blend. Tensile and tear strength and ozone resistance of the blend with 5 phr EPDM-g-GMA were better than conventional NR/BR blend of tire sidewall.

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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